

Smart Transducer Integrator (STI) for Standardized System-of-Systems Monitoring

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Abstract—With the proliferation of embedded devices and communication systems as well as their optimization in cost, size, and power requirements, at the moment we can find a large number of applications for Smart Sensor Networks (SS-Net) and Internet of Things (IoT) systems. Applications include transport systems (such as oil, water, and gas pipelines), smart houses, and wearables, just to mention a few. Needs when working with these types of systems are standardization, modularity, and scalability considering reducing costs associated with integration, upgrades, interoperability, portability, manageability, and the preparation required to design and configure new systems. Measurability, testability, and reliability are also important aspects that should be considered in the design of critical systems. In this paper, the Smart Transducer Integrator (STI) system is introduced, which provides a building block for the implementation of SS-Nets and IoT. The STI aims to respond to the current SS-Net and IoT needs. Innovative aspects of the STI system are embedded standardized mechanisms for enhanced reliability and ubiquitous operation considering the capability to operate even in completely sealed or inaccessible environments, such as metallic confinements.

Keywords—smart transducers, smart sensors, smart sensor networks, internet of things, wireless sensor networks, IEEE 1451.0, Health Electronic Data Sheets, HEDS, TEDS

I. INTRODUCTION

Wireless Sensor Networks (WSN) and Internet of Things (IoT) are strong current technological trends due to a maturation of computer science, communications, information technology, embedded systems, measurement techniques, and portable devices, where all of these are the foundation for both WSN and IoT. Integrated Circuits (IC) are being increasingly optimized for size, weight, power, and cost (SWAP-C). These SWAP-C optimized and hybrid devices allow developing sensing systems with embedded computational and storage capabilities, which opens new design possibilities, such as in algorithm deployment, as well as in manageability resources to enhance measurability, testability, and reliability of the sensing system. WSN have been present since the 1980s (known at the beginning as a Distributed Sensor Network), although the term WSN was used at the end of that decade [1]. On the other hand, the Internet of Things term was introduced about two decades after the first WSN.

WSN and IoT are related technologies, but with important differences. Referring to their definitions, a sensor network

consists of a large number of interconnected sensors' nodes with the capability to perform sensing and data collection. According to the IEEE Internet of Things Journal [2], an IoT system on the other hand is a network of networks, where typically a massive number of objects/things/sensors/devices are connected through communications and information infrastructure to provide value added services by utilization of intelligent data processing and management for different applications. The WSN original focus was the interconnection of machine-to-machine (M2M), where examples include the SCADA networks and the early highly customized control and instrumentation systems [1]. In the case of IoT, a wide set of features are enabled: (i) human-computer interaction (HCI); (ii) M2M interaction; and (iii) interactions of heterogeneous entities. Commercially speaking, an increase in revenues is expected in markets related to these technologies as more applications are being increasingly identified. The primary advantage of the Internet of Things is that it connects "things" any-time, any-place, for anything, where heterogeneous entities can be interlinked and then each physical entity has a virtual counterpart which is aware of its context and can communicate to exchange data, information, and knowledge to make informed decisions through these interrelationships. As we can see, IoT enables interaction among real/physical and the digital/virtual planes. WSN and IoT applications are numerous, and include structural health monitoring, automotive, energy, smart houses, health, agriculture, logistics, smart cities, and wearables, to mention just a few. WSN is a core technology of these kinds of applications as well as IoT systems (IoT builds upon WSN and cloud computing, but also is related to RFID and middleware).

Focusing to WSN, important advances had been made in the R&D community, but still efforts are needed when considering technology standards at different system levels such as safety, system operation, and hardware interfaces. The Smart Transducer Integrator presented in this paper looks to support these needs at the real/physical level when considering WSN and IoT type of applications. This technology is exemplified for its use within an enterprise health monitoring application moving toward a future IoT implementation.

The Smart Transducer Integrator (STI) is a new SWAP-C optimized and customizable framework with an optional base-line standardized communication stack. It is a multi-functional sensor node foundation whose primary purpose is to integrate (i) monitoring, diagnostics, and management capacity of systems-

of-systems through WSN realizations with (ii) enterprise type infrastructures. The STI is multi-functional since it also allows for integration of: (a) communication bridges (through advanced ultrasound interfaces [3], for example); (b) smart sensors (and when integrated with networking technologies, smart sensor nodes as well); and (c) raw data acquisition systems. Its maximum utility and performance is obtained when the STI operates in a network environment, with diagnostic algorithms (self-diagnostics and local system diagnostics), standardized communications, and once the WSN is formed, full integration into the Enterprise Infrastructure. The STI builds upon State-of-the-Art (SOTA) devices by integrating standard communication ports, baseline sensors, and advanced ultrasound communication schemes. It is designed to be reconfigurable through a set of flexible hardware and software schemes. Available resources for customization include embedded ultrasound communication, a standardized communication stack designed under the IEEE 1451.0 (Dot0) framework [4], and hardware reconfiguration schemes.

When the full STI system is deployed (i.e. when loading the communication stack), the on-line processing capability of the STI also allows for the execution of self-diagnostic algorithms and local system diagnostics under an innovative standardized framework termed Health Electronic Datasheets (HEDS), which builds upon the Dot0 standard. The STI inherits other advantages by developing the communication stack under the Dot0 framework, such as enhanced manageability (configuration management) through Transducer Electronic Datasheets (TEDS), standardized sets of commands (and reply messages), and a system/sensor failure awareness mechanism.

In this paper, we introduce the STI system for the application of distribution system monitoring under, as an example, the scope of NASA test systems enhanced reliability. As mentioned above, the system is general purpose and can be applied within other work scopes. The content of the paper is organized as follows. Part 2 describes the STI system, the system evolution, reconfiguration schemes, and standardized features. Part 3 then briefly details the innovative features of the STI system. Part 4 describes application to a current system development. Finally, Part 5 addresses conclusions and future work.

II. STI SYSTEM DESCRIPTION

A. System Evolution and Design Philosophy

A great deal of progress has been gained both for WSN and IoT. Advances that propelled the WSN development include: (a) sensor nodes size reduction; (b) implementation to different application domains; (c) data transmission range enhancement; (d) ability to embed more computational power; (e) low-power consumption optimization; and (f) communication reliability improvement. A large number of activities in the WSN scientific research and development community can be observed after 2000 [1]. Regarding IoT, the first publications appeared after 2000, and since 2010 they have increased significantly [1]. IoT is an emerging trend that arising from the internet and uses IPv6 to ensure that every “thing” has an identifiable Internet address. On the other hand, WSN can use the internet, but it is not required. Standardization efforts have also been conducted,

where the IEEE 1451 set of standards are especially noteworthy [5], where the standards were designed to be complimentary.

Within this context, American GNC Corporation (AGNC) has been involved in several government funded developments, which led to different software and sensor technologies that have since undergone further R&D at the company for improvements, resulting in a set of commercial products. AGNC’s STI product inherits the advanced characteristics of AGNC’s previous Smart Sensor products, namely the Autonomous Intelligent Sensor for Tracking Systems (AISTS) and the coremicro® Reconfigurable Embedded-Smart Sensor Node (CRE-SSN). The STI product evolution is depicted in Fig. 1, where main characteristics of the CRE-SSN [6] and AISTS [7] (foundations of the STI) are listed. CRE-SSN main characteristics are a standardized design (set of commands, reply messages, TEDS, and operational modes as per the Dot0 standard), low power, and customization capability through an expansion board. Looking for an optimized SWaP-C system with vibration signature recognition, the AISTS was designed for ultra-low power consumption, embedded adaptive pattern recognition by embedded artificial neural networks, and uses a state of the-art FRAM memory microcontroller (MCU). The STI blends all of these capabilities to obtain a customizable hardware and software framework to build WSN. Considering standardization, it should be noted that Dot0 does not include schemes for conducting smart sensor health monitoring, which is the area to which the novel HEDS implementation is focused, as addressed in Section III.

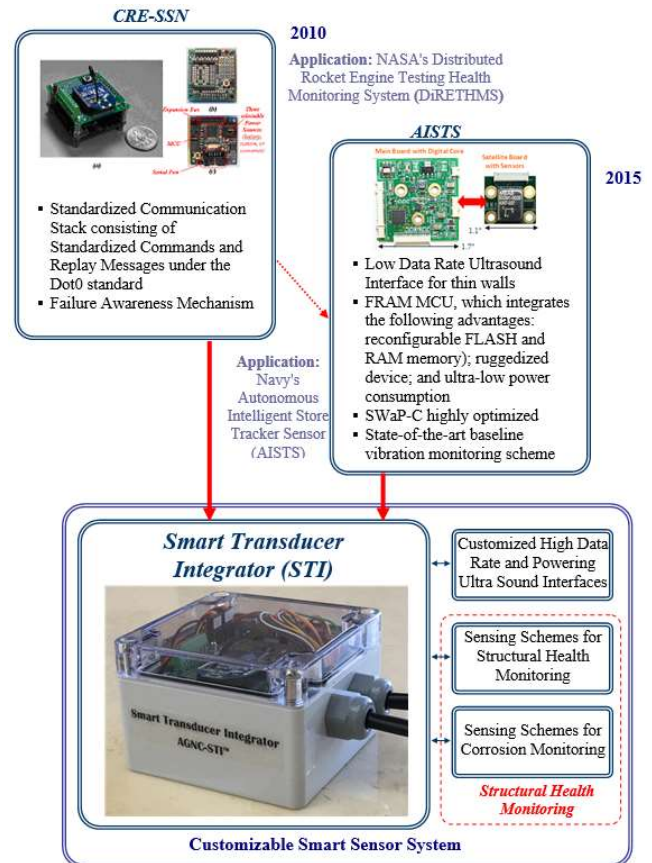


Fig. 1. AGNC's embedded systems with Optimized SWaP-C designs, exemplifying the STI system evolution

Smart Transducer Integrator (STI), Autonomous Intelligent Sensor for Tracking Systems (AISTS) and coremicro® Reconfigurable Embedded-Smart Sensor Node (CRE-SSN) are Copyrighted by AGNC

Fig. 1 also exemplifies the additional customization work to obtain a Smart Sensor capable of conducting Structural Health Monitoring (SHM) by providing the required Smart Sensor's structure at the bottom part of the figure. Considering this customization example, and looking at the figure, in addition to the STI, three additional blocks are required to obtain the full SHM smart sensor node. In this example, deploying the sensor in a metallic confined space is considered, where an ultrasound communication interface can be customized to handle high data rate communication (and power transfer) through the metallic wall without physical penetrations and according to the wall's characteristics (thickness and material). Sensor suites would include, e.g., corrosion and structural monitoring transducers to track signals correlated with the structure's health condition.

B. STI System Structure and Description

The STI, being reconfigurable, is designed to be a general-purpose system, and consists of a MPU board and an expansion board with connection capability to SOTA vibration monitoring satellite board (as a baseline, the legacy AISTS sensing scheme). Looking at the block diagram in Fig. 2, the STI's MPU board consists of: (a) an ultra-low power consumption processor with Ferromagnetic RAM (FRAM) technology [8] from Texas Instruments (depicted in the MCU block at the top left); (b) easy connections to a flexible array of standard ports (UART, SPI, and I2C); (c) cutting edge baseline ultrasound communication interfaces (from the internal AISTS and external customized interfaces); (d) embedded standalone temperature event detector (STED) within the STI digital core; (e) expansion bus; (f) power and carrier signal quality monitoring; and (g) powering circuits. Powering is enabled by many options, e.g. rechargeable battery, AC power, ultrasonic transmission, energy harvesting, etc.

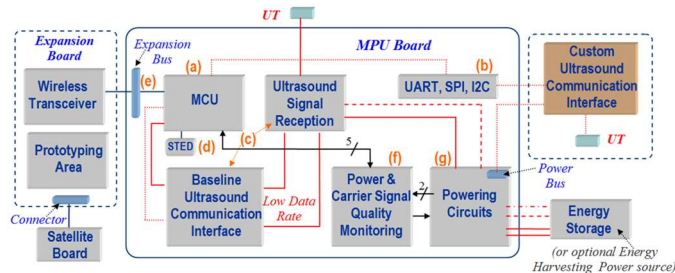


Fig. 2. STI Diagram (MPU board, expansion board, and satellite board)

a) *Energy Harvesting*: The STI has been designed for applications where operating at energy harvesting level power consumption is a desired capability. For example, a 3.7V 450 mAh Lithium-Ion Polymer Rechargeable Battery can provide charge to the STI sensor for anywhere between 28 days (always active) up to 364 days (always asleep). Since this type of battery cannot usually provide a discharge current at less than -20°C , in parallel, a supercapacitor can be used to operate the device for a few hours in the -40 to -20°C range in special applications.

b) *Baseline Sensors*: Since the STI has all the capabilities (and more) of the AISTS, it provides, as a baseline, a SOTA vibration monitoring sensing scheme through a satellite board (small block depicted at the bottom left in Fig. 2, and the picture shown in Fig. 1 at the right). This small and compact board has a very high performance triaxial analog accelerometer and an ultra-low power digital triaxial accelerometer. The primary high

performance accelerometer operates with a supply current of less than $22\text{ }\mu\text{A}$, wide bandwidth of up to 6000 Hz , and nominal sensitivity of 50 mV/g . The STI's baseline signal conditioning produces a very flat frequency response up to about 2 kHz , making it an excellent option for vibration monitoring within this operational bandwidth. The ultra-low power digital triaxial accelerometer allows awaking the MPU when vibration samples exhibit a behavior correlated with an event of interest (activity can be an impact, for example), operates in a standalone way to conserve power ($23\text{ }\mu\text{A}$ in measurement mode), and contains a rich set of features (tap detection, activity monitoring, and free-fall detection, among others).

c) *Modularity and Scalability*: System reconfigurability is enhanced by a modular design where customized expansion boards can be attached to the MPU board through an expansion bus, which provides interfacing signals to tailor the Transducer Interface Modules (TIM) to meet specific requirements in target applications. Available signals in the expansion bus include the MPU's ADC channels (termed "Transducer Channels" within the Dot0 standard framework), GPIO, and powering.

C. Reconfigurability

Hardware and software reconfiguration schemes are available for setting the STI either as: (a) wireless sensor (e.g. using ZigBee transceivers); (b) ultrasound based system [3] for operation in confined/inaccessible spaces; or (c) wired sensor. Fig. 3 and Fig. 4 describe available STI hardware configuration options, where powering methods are depicted in Fig. 3, and communication options are defined in Fig. 4. Regarding the powering methods, the STI has four options: (i) battery (where no ultrasonic power transmission is enabled); (ii) external power unit (such as a custom energy harvesting module or an AC/DC converter); (iii) ultrasound and rechargeable battery; and (iv) ultrasound and supercapacitor. A powering setting example is the case when energy is transferred through the ultrasound interface, as described in Fig. 3. In this powering case, the red solid line depicts the energy flow when the STI is configured in this operation mode. Ultrasonic waves are received and converted to electric signals by the Ultrasound Transducer (UT, consisting of a piezoelectric element), which are conditioned by the Ultrasound Signal Reception block. In this configuration, a battery charger is enabled, which then feeds a rechargeable battery. The output of the rechargeable battery (nom. 1.8 V) is fed to a charge pump to deliver a regulated voltage (3.3 V) to the STI system. Testability is achieved by monitoring the signals VBAT (output voltage in the battery), DC_UNREG (ultrasound signal voltage), and output battery charger status signals (FAULT and CHRG) in this powering operation mode.

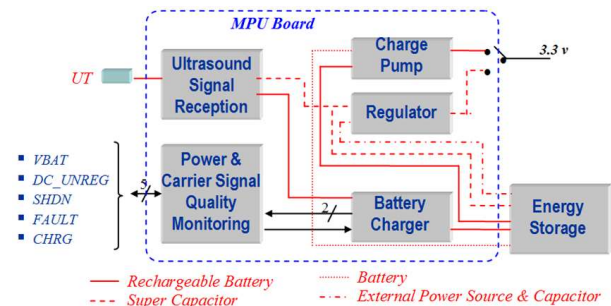


Fig. 3. STI's MPU Board Powering Scheme Options

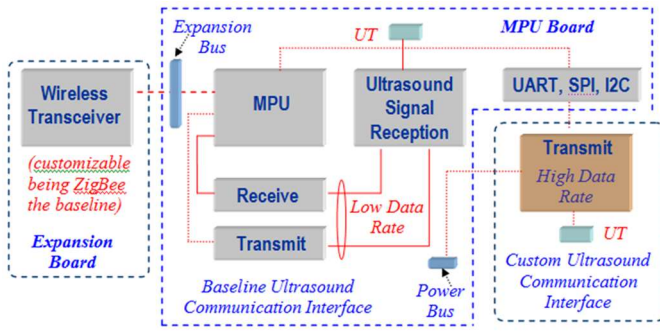


Fig. 4. STI Communication Options

Fig. 4 depicts the communication options, where the baseline wireless option is a ZigBee transceiver, which can operate with the STI's Dot0 standardized communication stack. In addition, wired communications are available through standardized ports (e.g. UART, SPI, and I2C). Considering enclosed areas or cases where wireless (RF) or wired communication are not desired, through wall communication by ultrasound techniques is available that maintains the integrity of the structure (no physical penetrations are used). The STI baseline includes ultrasound communication interfaces for thin walls and low rate data transferal (2.3 kb/s approximately). In the case of thick walls, higher data rates (e.g. 100 kb/s to 1 Mbps), or if different types of structure materials are targeted, the STI allows for ultrasound interfaces customization. This is the case exemplified in Fig. 4, where a high data rate transmission module can be integrated to achieve this. Thus, the STI allows designing many core communication interfaces (where ultrasound energy transferal is also an option). The STI can extend networking capabilities in difficult to access metallic environments (such as in ships and airframes).

Complementing the hardware reconfiguration schemes, the STI also includes a software component based on the optional Dot0 communication software stack. By working under Dot0 formalisms, additional benefits are: (a) self-identification; (b) calibration and self-diagnostic support; and (c) communication protocol. Within the functionalities as defined by the Dot0 standard, sets of Transducer Interface Modules (**TIM**) interact with a Network Capable Application Processor (**NCAP**) [4]. Some remarks are as follows:

- For compliance with Dot0, TIMs contain a set of n_s Transducer Channels, where each Transducer Channel "x" (TCh_x) is associated with a physical transducer (sensing element or actuator).
- The Dot0 standard also defines Transducer Electronic Datasheets (**TEDS**) with the purpose to represent in an electronic way (tables stored in the device's nonvolatile memory): transducer specifications (e.g. types, available sampling modes, sampling period, resolution, scale, units), transducer name, manufacturer, and calibration information. Three TEDS are associated with each TCh_x at a minimum (defined by Dot0) and three TEDS are associated to the TIM (to define overall system operation and communication specifications).
- Each attached transducer (sensor or actuator) will be associated to an individual set of TEDS consisting of:

User's Transducer Name TEDS, Calibration TEDS, and Transducer Channel TEDS.

- Overall attributes of the TIM (smart sensor) are defined by a set of TEDS consisting of Meta-TEDS (contains a Universal Unique Identification (UUID)), PHY-TEDS, and User's Transducer Name TEDS.
- Dot0 provides a set of commands and associated reply messages to: (i) **drive the sensor operation** and (ii) **manage TEDS** (read, write, and modify). This Dot0 capability supports Configuration Management (to identify sensors).
- TEDS access is obtained through the Dot0 set of commands. After reading the TCh_x 's TEDS set (for example, during the TIM's initialization process) the specifications of the corresponding sensor are known. *Under this scheme, the Dot0 provides self-identification capability for the TIM and each attached sensor.*

III. EMBEDDED HEALTH ELECTRONIC DATASHEETS (HEDS)

When deploying the communication stack, a new innovative STI capability is the infusion of Health Electronic Data Sheets (HEDS), which have been developed with the purpose of providing a standardized sensor self-diagnostic framework within smart sensors. Initial concepts of HEDS for the health of transducers began with D. Jethwa, et. al. in 2008 [9] and R.L. Oostdyk, et. al. in 2006 [10], and have since been expanded and enhanced in our recent work from [11] leading to the results and standardized scheme described and proposed in reference [12], which is summarized in this section.

A goal of the proposed standardized framework is to enable instantiation processes to perform sensor self-diagnostics. To achieve this, a special Dot0 Transducer Electronic Data Sheet (TEDS) has been proposed. We named this special TEDS the Health Electronic Data Sheet (HEDS) because it contains a set of parameter values for involved diagnostic algorithms. This HEDS definition follows the Dot0 - 2007 TEDS conventions. There is an univocal correspondence between a given algorithm and its parameters as stored in the HEDS. In this way, self-diagnostics can be readily conducted by: (i) reading the HEDS content; (ii) using the stored parameters to instantiate a new process (program), which are part of this HEDS definition; and (iii) utilizing the active process (program) to test a sensor fault condition. Algorithms considered for the initial HEDS definition fall into two categories: (1) data centric and (2) system centric algorithms. Data centric methods detect faults using a sensor's own data. Cases accommodated in our HEDS framework include: (a) outliers; (b) spikes and (c) variance (stuck or high noise). System centric methods require analysis of multiple sensors, where cases included are offset, gain and drift, etc.

A special register termed the HEDS Status Control (HEDS-SC) register provides control over the execution of each sensor diagnostic algorithm and tracking the TIM's health status. The HEDS-SC is an extension to the Dot0 standard (i.e. this register is not defined by the Dot0 standard). By working under the Dot0 standard formalism, the HEDS framework inherits the following advantages: (1) standardized commands are available to interact

with it (Query, Read, Write, Update, Self-Test); (2) efforts for standardization are expedited; and (3) HEDS may have a greater users' acceptance. Dot0 does not include schemes for conducting smart sensor health monitoring, which is the area to which this HEDS framework is focused. Since the HEDS is a special TEDS case, all Dot0 definitions apply to the proposed structure. Detailed HEDS structure and conformation definition, HEDS-SC register description, diagnostic schemes, and an example of the current HEDS version can be found in [11][12].

IV. STI APPLICATION CASE

Because the STI is reconfigurable, it can be applied for a broad variety of SS-Net applications. With only some additional customization work, it has the potential to be utilized in the development of IoT type systems. At the moment, there are three current target applications of the STI, where sensor suites and custom interfaces will be added to obtain system versions for. These include: (a) through thick metal wall communication and fiber optic networking environments; (b) structural health monitoring in confined spaces; and (c) distribution systems. In this section, the STI application is described within the scope of distribution systems, which is a current NASA development.

A. STI application to the Integrated Monitoring Awareness Environment (IM-AWARE) System

In this application, the STI is being customized to operate as wireless/wired TIM, where a ZigBee module provides wireless communication capability to the networked processors, e.g., the Dot0's NCAP. However, it is also designed to integrate IEEE 802.11 based modules to leverage IP networking capabilities.

The Integrated Monitoring AWAREness Environment (IM-AWARE) is a networked health monitoring system that is under development for a NASA STTR project [11]. The IM-AWARE system architecture shown in Fig. 5 comprises an Enterprise Infrastructure of Smart Sensor Networks (SS-Nets), a web based server containing a database for managing data, information and knowledge (DIAK), and remote clients (which can be devices like smartphones) providing health monitoring, real-time DIAK, and maintenance guidance capabilities.

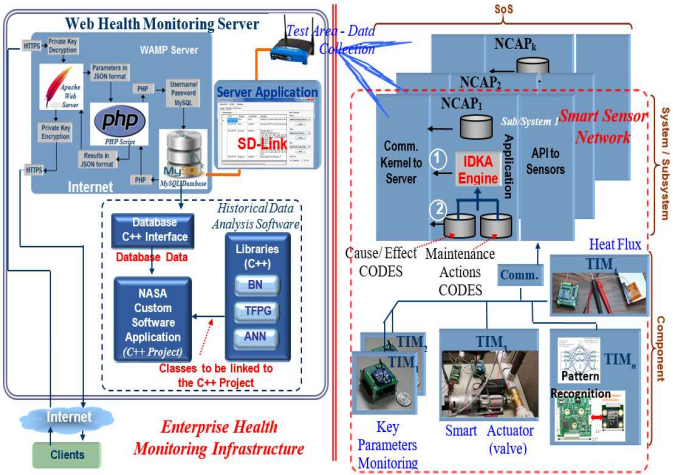


Fig. 5. System-of-Systems Monitoring through Enterprise Health Monitoring Infrastructure (EHMI) and Smart Sensor Network (SS-Net) System

Referring to Fig. 5, within the IM-AWARE system's overall architecture (on the left side of the central vertical red line), an Enterprise Health Monitoring Infrastructure (EHMI) is closely coupled with standardized SS-Nets. The SS-Nets provide a distributed processing platform for execution of system centric diagnostic algorithms. Then, at the sensor level, data centric algorithms are deployed in the customized STI by using HEDS as summarized in Section III. The IM-AWARE provides a highly distributed architecture suitable for System of Systems (SoS) monitoring, where TIMs forming a SS-Net can be deployed within a distribution system and through the EHMI to provide health information to users through smart phones, tablets, and PC type computers. This capability can be scaled to several systems, where a set of SS-Nets can be deployed among several systems, and in this way track the SoS condition. Because of the STI local processing ability, advanced custom diagnostics are being developed as described in the following paragraphs considering leakage detection.

B. Advanced Features: Diagnostics (Leakage)

The STI has been designed for deployment as an intelligent node for embedded sensing and analysis in health monitoring applications. One such case is vibration signature monitoring of structures, machinery, and even the presence of leaks in fluid pipe networks as detailed in this section. For evaluating leakage detection, the testbed illustrated in Fig. 6 below was fabricated consisting of a motor pump, long copper pipeline, plastic pipes around the pump, and a water basin. Small holes in the copper pipe are covered by adjustable rubber rings that allow to produce leak events. The goal is to utilize the STI's expansion board and satellite board (see Part II.B) with accelerometers mounted to the pipe's outside for measuring and acquiring vibrations, and the STI's main board for conducting data acquisition and on-line diagnostic analysis of vibrations for leak identification. From Fig. 6, the photo at the left depicts the entire testbed, the lower right shows the expansion board with a triaxial accelerometer, and the upper right is of the main board. For this photo, AGNC's AISTS was used as a smart sensor, and data acquisition and processing took place in a PC; however, the STI has the capability to do all of this as a small powerful sensor node in a fully embedded way.

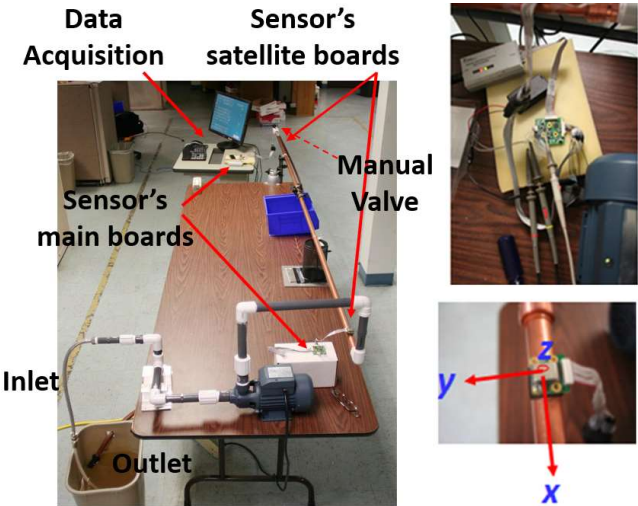


Fig. 6. Leakage Identification Testbed with Smart Sensor Node

The STI can perform data acquisition and save digital data in the MPU's memory for data characterization efforts. For this case, after acquiring data for no, small, and medium leaks, time domain data sequences showed little/no observable distinctions; however, windowed FFTs made the differences readily evident. This is illustrated by the accumulated spectra in Fig. 7 for each case, where frequencies and spectral densities in the highlighted low frequency band are different. The main MPU board enables configurable sample rates and selection of sensor channels for tailoring the data acquisition process to best accentuate the leak characteristics, as well as capability of on-line frequency domain analysis (which provides a better indicator of leak presence).

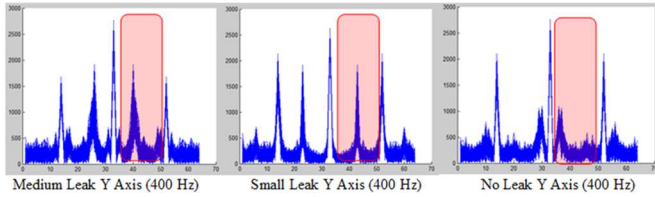


Fig. 7. Frequency Analysis of Vibration Data for Leakage Detection

The next step was to design the leak identification algorithm to deploy in the STI main board's MPU for in-situ diagnostics. AGNC's vibration analysis and neural network design software suite was used for designing a leak detection scheme. This suite enables exploring statistical features, optimizing neural network (NN) classifier performance by finding feature sets to minimize classification error, and designing networks for classification. It allows computing statistical features of time or frequency data as inputs into a NN including: RMS, mean, variance, skewness, kurtosis, crest factor, min, max, gradient, selected frequencies, etc. For this case, the feature set of y axis: RMS, mean, and skew was discovered to yield the smallest classification error ($< 1\%$) while having low computational burden. Fig. 8 depicts the mean feature signature at the top, results of NN training at the lower left, and the current leakage classification at the lower right.



Fig. 8. Vibration Analysis and Neural Network Design Software Suite

Finally, the trained neural network and selected feature set is then transferred to the MPU of the STI main board (over the STI communication lines). For online operation, the MPU processes incoming vibration signatures (by conversion into the frequency domain via windowed FFTs), extracts features, feeds them into the neural network, and produces an in-situ leak classification.

V. FINAL COMMENTS AND CONCLUSIONS

An innovative and flexible hardware and software scheme has been presented, resulting in the Smart Transducer Integrator product, which allows for integration of diverse sensing schemes and many different basic communications, but also standardized smart transducers with advanced communication interfaces, e.g. ultrasound. One of the main goals of this product development was to provide a hardware and software sensing infrastructure that allows hosting innovative HEDS. Sensor self-diagnostics and system diagnostics within a standardized framework is a system advantage and strategy strongly pursued through the STI development. The STI is both a platform and foundation, which is currently being integrated in several ongoing R&D efforts as exemplified in this paper.

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